

A Meta-Analysis of Deep Learning and Agentic Artificial Intelligence for Forecasting Cyberattacks in Educational Institutions

Thushan Amarasinghege^{1[0009-0006-1104-5951]*} and Kalpdrum Passi^{1[0000-0002-7155-7901]}

¹ Laurentian University, Sudbury, Ontario, Canada
{tamarasinghege, kpassi}@laurentian.ca

Abstract. This research paper investigates the use of Deep Learning and Agentic Artificial Intelligence (AI) on forecasting possible cyberattacks on educational institutes like schools and universities. Due to different user groups and limited security resources, educational institutions frequently face cybersecurity threats, and thereby the need for forecasting cyberattacks and adaptable security mechanisms. In this paper, a meta-analysis is presented from research studies conducted from year 2015 to 2025 that are based on AI to maintain the security of schools and university networks. The goal is to conduct an assessment comparison between the previous studies by using standard performance metrics and the operations handled by independent agents. This analysis concludes that higher efficiency can be achieved via Agentic AI in terms of higher adaptability and higher efficiency of responses. Furthermore, the ability to identify the patterns through deep learning is successfully achieved. This meta-analysis can benefit securing the educational institutes.

Keywords: deep learning; agentic artificial intelligence; cybersecurity, educational infrastructure.

1. Introduction

Educational institutions represent a high-stakes environment for cybersecurity due to their heterogeneous user populations and the expansive adoption of "Bring Your Own Device" (BYOD) policies. Unlike corporate environments with predictable traffic, academic networks face irregular, semester-based surges and often operate with limited security resources. This research presents a meta-analysis of studies from 2015 to 2025, evaluating the transition from passive pattern recognition to autonomous, agent-based mitigation strategies.

2. Background and Related Work

The research compares two different ways of using AI to protect these networks:

Deep Learning (The Pattern Finder): This is great at spotting known attack signs in massive amounts of data. It's reliable (87% accuracy) but reactive, it mostly just flags a problem and waits for a human to fix it.

Agentic AI (The Independent Responder): Think of these as autonomous digital security guards. They don't just watch; they "reason" and take action, like shutting down a compromised server before the damage spreads. This approach hits about 92% accuracy.

Methodology and Comparative Results

The meta-analysis synthesized data from academic databases (IEEE Xplore, Scopus) using a **random-effects model** to calculate weighted mean accuracy across diverse studies.

To make sure the final numbers weren't skewed by smaller, less reliable studies, the researchers used a **Random-Effects Model**. Think of it as a "fairness filter" that gives more weight to large-scale experiments while accounting for the fact that every school network is a little different.

$$\bar{x} = \frac{\text{sum of } \textit{Weights} * \textit{Value}}{\text{sum of } \textit{Weights}} \quad (1)$$

$$\bar{x} = \frac{\sum_{i=1}^n \omega_i x_i}{\sum_{i=1}^n \omega_i} \quad (2)$$

$\bar{x}$ is the weighted mean accuracy

x_i is the accuracy reported in study i .

ω_i is the weight (inverse of the variance) for study i . This ensures that studies with larger sample sizes have a proportional impact on the results of the meta-analysis.

Table 1. Aggregated Performance Metrics of Evaluated Artificial Intelligence Paradigms.

AI Paradigm	Mean Accuracy	Mean Precision	Key Strength	Primary Limita- tion
Deep Learning	87%	84%	Pattern recog- nition [5]	Dependence on labelled data
Agentic Artificial In- telligence	92%	92%	Autonomous adaptation	Governance complexity

Why Speed Matters: In a cyberattack, every minute counts. The study found that switching to Agentic AI can cut the Mean Time to Respond (MTTR) in half.

- **Before:** It took an average of **80 minutes** to deal with a threat.
- **After:** That time dropped to **40 minutes**.

A Smarter Way to Deploy: The 4-Stage Pipeline: The study proposes a four-stage pipeline tailored for academic digital infrastructure:

1. **Data Input:** Ingesting raw network traffic including login attempts and cloud access.
2. **Pattern Recognition:** Leveraging Deep Learning to establish a "Normal Behavioral Pattern".
3. **Anomaly Detection:** Utilizing Generative AI to identify outliers and simulate adversarial scenarios.
4. **Autonomous Response:** Deploying Agentic AI to isolate servers or block threats in real-time without manual intervention.

Conclusion and Future Directions

While Agentic AI offers superior adaptability, its governance complexity requires human-in-the-loop safeguards to prevent unintended service disruptions. Future work will investigate **Federated AI** to preserve data privacy and the development of standardized benchmarking for autonomous security agents in live operational environments.

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